

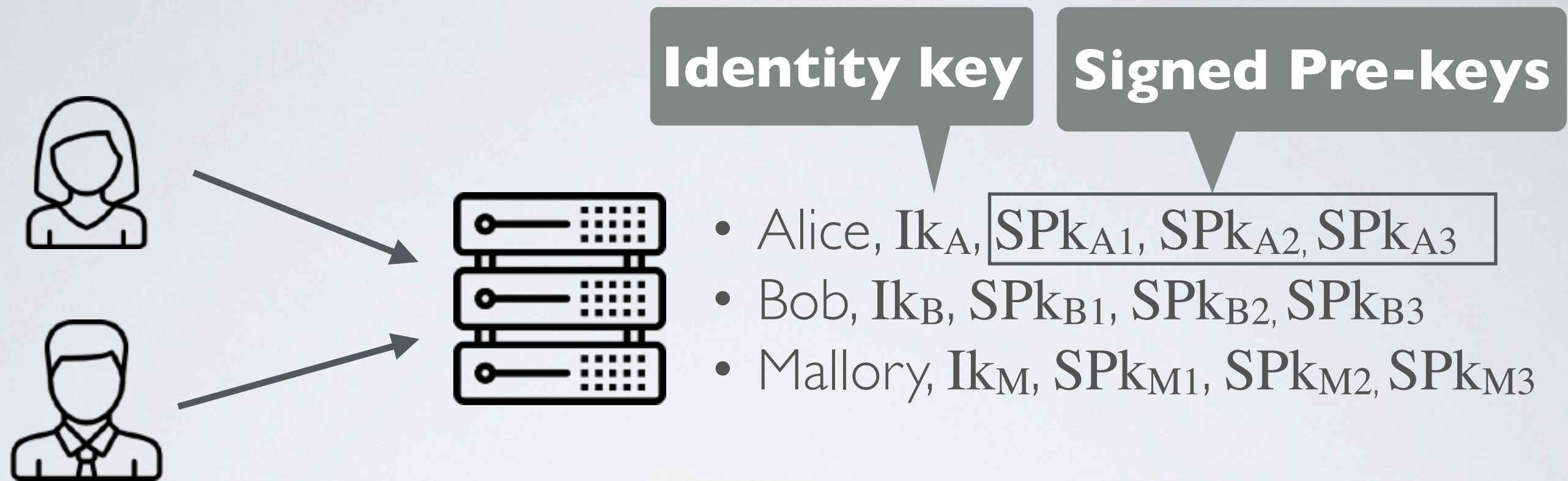
Signal in a Nutshell

➡ Asynchronous Protocol

Two phases

- **Triple Diffie-Hellman Key Exchange (3DH)**
Open a communication channel between Alice and Bob
- **Double Ratchet Protocol**
Exchange message exchange with forward secrecy

Signing-up with Signal



- Users sign-up with the signal server that verifies their identity (email, phone number, 2FA and so on)
- Users upload an identity key (public) and several pre-keys (public) signed with their identify key